B.Tech. Degree VI Semester Special Supplementary Examination November 2012

ME 606 CAD/CAM
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Explain the various types of geometric modelling.

(b) Write a note on Detroite type of automation.

(c) Write a note on canned cycles.

(d) Describe the advantages and limitations of CNC.

(e) Explain the term 'CAPP'.

(f) Write short notes on control systems of CNC.

(g) Describe the concept of FMS.

(h) Discuss on the end effectors in robots.

PART B

(4 × 15 = 60)

II. Explain the various methods by which the data is exchanged between CAD and CAM. Also mention each of its applications, advantages and limitations.

OR

III. (a) What is line balancing?

(b) Explain the various kinds of production systems with reference to its level of automation.

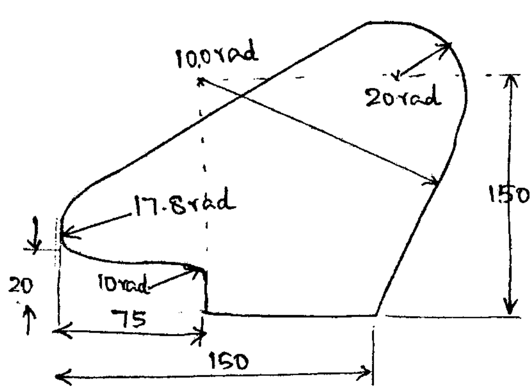
IV. (a) Compare a CNC system with a DNC system.

(b) What are the various classifications of CNC machines?

OR

V. (a) Write notes on APT.

(b) Write the APT geometry statements to derive the outline of the cam shown in the figure.



VI. Discuss the features of CNC systems in detail.

OR

VII. Explain the method of testing of NC machine tools. Also suggest methods to remove its static and dynamic errors.

VIII. Explain the various elements of a CIM system in detail.

OR

IX. With neat sketches, describe the common types of robot configurations.

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